

IoT-Based Autonomous Weed Detection & Removal Robot

Master Project Execution Plan

Current Status (Estimated from Repository & Discussion)

- Backend and frontend structure created (Node.js/Express based project).
- ESP32-based robot control architecture selected.
- Phase 4 hardware planning completed.
- HP Victus (i5 14th Gen, RTX 3050, 16GB RAM) selected for AI processing.
- Raspberry Pi, Orange Pi, Jetson Nano removed from architecture.
- Wireless camera streaming architecture selected.

Final System Architecture

LILYGO T-SIMCAM (OV5640 5MP Camera) → Wi-Fi Hotspot → HP Victus Laptop (YOLOv8) → Wi-Fi → ESP32 DevKit V1 → Motors, Cutter, Sensors.

Completed / Mostly Completed

- Project structure.
- Backend foundation.
- Frontend foundation.
- ESP32 selection.
- Phase planning.
- Component research.
- Robot control architecture.

Pending Phase 4 Tasks

1. Assemble 4WD chassis.
2. Install motors and wheels.
3. Connect 2× L298N motor drivers.
4. Connect RS775 cutter motor through relay.
5. Connect HC-SR04 sensors.
6. Configure ESP32 motor control firmware.
7. Create movement APIs.
8. Create control dashboard.
9. End-to-end testing.
10. Safety testing and emergency stop.

Phase 5 Tasks (AI Integration)

1. Install LILYGO T-SIMCAM.
2. Stream camera feed via Wi-Fi.
3. Install Python, OpenCV, Ultralytics YOLOv8.
4. Create weed dataset.
5. Train YOLO model.
6. Detect weeds in real time.
7. Send commands to ESP32.
8. Automate cutter activation.
9. Field testing and tuning.

Team Allocation (4 Members)

Member 1 – Backend APIs & Database.

Member 2 – Frontend Dashboard.

Member 3 – ESP32 Firmware & Communication.

Member 4 – Hardware Assembly, Testing & Documentation.

Required Hardware

- ESP32 DevKit V1
- LILYGO T-SIMCAM ESP32-S3 + OV5640
- 4WD Robot Chassis
- 4 DC Gear Motors
- 2× L298N Motor Drivers
- RS775 High Torque Motor
- Relay Module
- HC-SR04 Sensors (2)
- 12V Battery Pack
- LM2596 Buck Converter
- Emergency Stop Switch
- Wiring, Nuts, Bolts, Brackets

Software To Install

Laptop:

- Python 3.11
- VS Code
- Git
- OpenCV
- Ultralytics YOLOv8
- Flask or FastAPI
- MQTT Explorer (optional)
- Arduino IDE

ESP32:

- ESP32 Board Package
- WiFi Libraries
- Camera Libraries

Project Deliverables

Phase 4 Deliverable:

Robot movement + cutter control + obstacle avoidance.

Phase 5 Deliverable:

Autonomous weed detection and cutting using YOLO.

Risks & Recommendations

- Test camera streaming before training AI.
- Use local hotspot only; internet is not required.
- Collect your own weed dataset for best accuracy.
- Mount camera 20–30 cm above ground and angle slightly downward.
- Keep AI processing on laptop; avoid adding Raspberry Pi.